

CAMBRIDGE INTERNATIONAL A LEVEL CHEMISTRY (9701)

Topic 26 — Reaction Kinetics

Paper 4 Topical Examination Pack & Comprehensive Examiner Mark Scheme

CANDIDATE	Urwah
SYLLABUS	Cambridge International A Level Chemistry (9701) — Paper 4
TOPIC COVERAGE	Rate Equations · Orders of Reaction · Rate Constants · Half-Life · Multi-Step Mechanisms · Arrhenius Equation
COMPONENT	Paper 4 — A Level Structured Questions (Theory & Calculations)
TOTAL MARKS	220 Marks (50 Comprehensive Questions)

HOW TO USE THIS A LEVEL PAPER 4 PACK

1. Complete each structured question thoroughly in the answer spaces provided, adhering strictly to mark allocations.
2. Questions are classified into Core and Advanced tiers, covering multi-step chemical derivations, mechanisms, and numerical proofs.
3. Every question features authentic Cambridge A Level Paper 4 examination references (e.g. 9701/42/M/J/23) for benchmark cross-referencing.
4. Full point-by-point examiner mark schemes with step-by-step working and error-carried-forward (ECF) criteria are detailed at the end.

SYLLABUS SUBTOPICS COVERED

26.1 Rate Equations & Orders of Reaction

— Definition of reaction rate, order of reaction (0th, 1st, 2nd), overall order; rate equations and rate constants

26.2 Continuous Monitoring & Half-Life Kinetics

— Continuous experimental techniques (sampling/quenching, colorimetry, gas collection, conductivity)

26.3 Reaction Mechanisms & Rate-Determining Step

— Multi-step reaction profiles; molecularity of elementary steps; identifying the rate-determining step

26.4 Arrhenius Equation & Activation Energy

— Effect of temperature on the rate constant k ; Arrhenius equation $k = A \exp(-E_a / RT)$; graphical determination of E_a

High-Frequency Core Repeats (Q41–Q50)

— The 10 most frequently tested Cambridge Paper 4 questions on Reaction Kinetics from the past 10 years

SECTION A: 6-MARK EXTENDED EXAM QUESTIONS (40% TARIFF · Q1–Q16)

1 Arrhenius Plot Determination of Activation Energy & Frequency Factor — 9701/42/M/J/23/Q3 [6 Marks]

HARD · Syllabus 26.4

The thermal decomposition of dinitrogen pentoxide, $2\text{N}_2\text{O}_5(\text{g}) \rightarrow 4\text{NO}_2(\text{g}) + \text{O}_2(\text{g})$, was studied at different temperatures. The rate constant k was measured, and an Arrhenius plot of $\ln k$ versus $(1/T)$ is shown in Fig. 1.1.

Data: Gas constant $R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$.

Coordinates on best-fit line: $(3.10 \times 10^{-3} \text{ K}^{-1}, -5.97)$ and $(3.40 \times 10^{-3} \text{ K}^{-1}, -8.67)$.

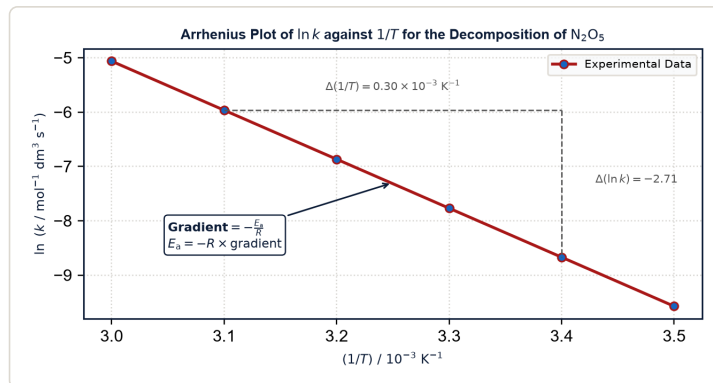


Fig. 1.1: Arrhenius plot of $\ln k$ against $1/T$ for the decomposition of N_2O_5 .

- (a) Calculate the gradient of the line in Fig. 1.1, stating its sign and appropriate units. [2]

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- (b) Use your gradient to calculate the activation energy, E_a , of the reaction in kJ mol^{-1} . [2]

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- (c) Calculate the value of the pre-exponential factor, A , and explain the physical significance of A . [2]

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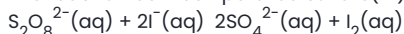
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2 Initial Rates Method & Deduced Rate Equation for Peroxodisulfate–Iodide — 9701/41/M/J/23/Q3 [6 Marks]

HARD · Syllabus 26.1

The reaction between peroxodisulfate(VI) ions and iodide ions was investigated:



Initial rates data at 298 K:

- Expt 1: $[\text{S}_2\text{O}_8^{2-}] = 0.040$, $[\text{I}^-] = 0.020$, Rate = $2.40 \times 10^{-5} \text{ mol dm}^{-3} \text{ s}^{-1}$
- Expt 2: $[\text{S}_2\text{O}_8^{2-}] = 0.080$, $[\text{I}^-] = 0.020$, Rate = $4.80 \times 10^{-5} \text{ mol dm}^{-3} \text{ s}^{-1}$
- Expt 3: $[\text{S}_2\text{O}_8^{2-}] = 0.080$, $[\text{I}^-] = 0.060$, Rate = $1.44 \times 10^{-4} \text{ mol dm}^{-3} \text{ s}^{-1}$

- (a) Deduce the order of reaction with respect to $\text{S}_2\text{O}_8^{2-}$ and with respect to I^- , showing your reasoning. [3]

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- (b) Write the overall rate equation and calculate the rate constant, k , including its units. [2]

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- (c) $\text{Fe}^{3+}(\text{aq})$ ions act as a homogeneous catalyst for this reaction. Write two equations to explain how Fe^{3+} catalyses this reaction. [1]

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3 Half-Life Analysis & Mechanism of Tertiary Bromoalkane Hydrolysis — 9701/42/O/N/23/Q3 [6 Marks]

HARD · Syllabus 26.2

The hydrolysis of 2-bromo-2-methylpropane, $(\text{CH}_3)_3\text{CBr}$, by aqueous hydroxide ions was monitored by measuring $[(\text{CH}_3)_3\text{CBr}]$ as a function of time.

The decay curve is illustrated in Fig. 3.1.

Successive half-lives measured from the curve are: $t_{1/2(1)} = 48 \text{ s}$, $t_{1/2(2)} = 49 \text{ s}$, $t_{1/2(3)} = 48 \text{ s}$.

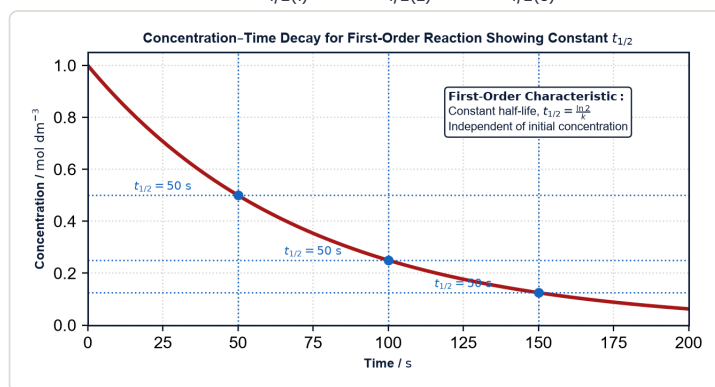


Fig. 3.1: Concentration-time decay curve for the hydrolysis of $(\text{CH}_3)_3\text{CBr}$.

- (a) Explain how the constant half-life demonstrates that the reaction is first-order with respect to $(\text{CH}_3)_3\text{CBr}$. [2]

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- (b) Calculate the value of the first-order rate constant, k , stating its units. [2]

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- (c) Doubling $[\text{OH}^-]$ has no effect on the rate. Write the full rate equation and outline a two-step $\text{S}_{\text{N}}1$ mechanism consistent with this finding. [2]

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4 Multi-Step Reaction Coordinate Energy Profile & RDS — 9701/41/O/N/23/Q3 [6 Marks]

HARD · Syllabus 26.3

The potential energy profile for a two-step gas-phase reaction, $\text{A} + \text{B} \rightarrow \text{C} + \text{D}$, is depicted in Fig. 4.1.

Step 1: $\text{A} + \text{B} \rightarrow \text{I}$ (Activation energy $E_{\text{a1}} = 65 \text{ kJ mol}^{-1}$)

Step 2: $\text{I} \rightarrow \text{C} + \text{D}$ (Activation energy $E_{\text{a2}} = 20 \text{ kJ mol}^{-1}$)

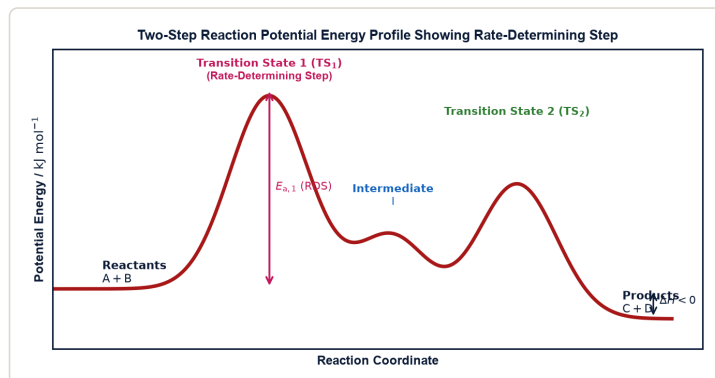


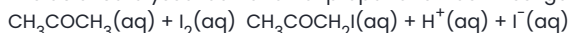
Fig. 4.1: Reaction profile illustrating transition states, intermediate I, and rate-determining step.

- (a) Identify the rate-determining step (RDS), and justify your choice with reference to the activation energies in Fig. 4.1. [2]
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- (b) Deduce the rate equation for this reaction mechanism. [2]
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- (c) State the chemical difference between an intermediate and a transition state. [2]
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5 Colorimetric Monitoring of Iodine-Propanone Kinetics — 9701/42/M/J/22/Q2 [6 Marks]

HARD · Syllabus 26.2

The acid-catalysed iodination of propanone was investigated:



The reaction was monitored using a colorimeter measuring the absorbance of brown iodine at 450 nm, as shown in Fig. 5.1.

The rate equation is known to be: $\text{Rate} = k[\text{CH}_3\text{COCH}_3][\text{H}^+][\text{I}_2]^0$.

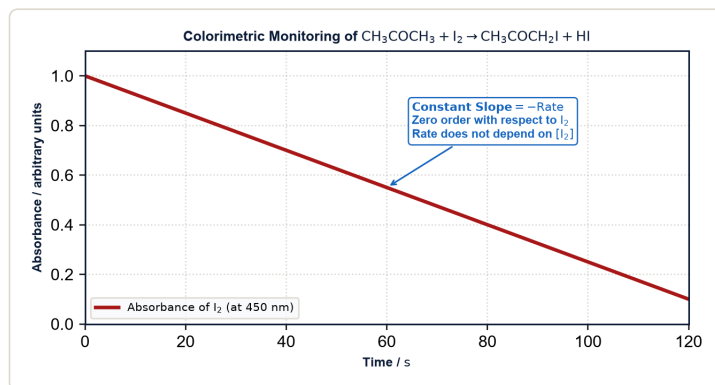


Fig. 5.1: Absorbance versus time showing linear consumption of iodine.

- (a) Explain how the linear graph in Fig. 5.1 confirms that the reaction is zero-order with respect to iodine. [2]
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- (b) Describe how the gradient of the line is used to calculate the rate of the reaction. [2]
- (c) Explain why H^+ appears in the rate equation even though it is not consumed in the overall reaction, and propose a mechanism step where H^+ is involved. [2]

6 Rate-Concentration Graphs & Rate Constant Units — 9701/41/M/J/22/Q2 [6 Marks]

HARD · Syllabus 26.1

The rate of three different reactions was plotted against the concentration of reactant A, as illustrated in Fig. 6.1.

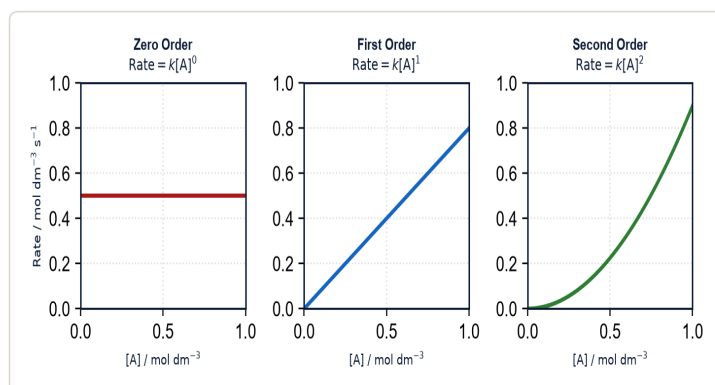


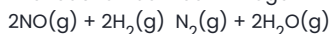
Fig. 6.1. Rate against concentration graphs for zero-order, first-order, and second-order reactions.

- (a) Match each graph (Graph 1: horizontal line, Graph 2: straight line through origin, Graph 3: upward curve) to its reaction order with respect to A. [3]
- (b) Deduce the units of the rate constant k for each of the three orders. [3]

7 Kinetics of NO and H₂ Reaction: Initial Rates & Rate Mechanism — 9701/42/O/N/22/Q2 [6 Marks]

HARD · Syllabus 26.1

The reaction between nitrogen monoxide and hydrogen gas was studied at 1000 K:



Experimental initial rates data:

- Expt 1: $[NO] = 0.0020$, $[H_2] = 0.0010$, Initial Rate = $3.00 \times 10^{-5} \text{ mol dm}^{-3} \text{ s}^{-1}$
- Expt 2: $[NO] = 0.0040$, $[H_2] = 0.0010$, Initial Rate = $1.20 \times 10^{-4} \text{ mol dm}^{-3} \text{ s}^{-1}$
- Expt 3: $[NO] = 0.0040$, $[H_2] = 0.0020$, Initial Rate = $2.40 \times 10^{-4} \text{ mol dm}^{-3} \text{ s}^{-1}$

- (a) Determine the order of reaction with respect to NO and with respect to H₂. [2]

- (b) Write the rate equation and calculate the rate constant k at 1000 K with its units. [2]

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- (c) A proposed mechanism is: [2]

Step 1 (fast): $2\text{NO} + \text{N}_2\text{O}_2$

Step 2 (slow): $\text{N}_2\text{O}_2 + \text{H}_2 \rightarrow \text{N}_2\text{O} + \text{H}_2\text{O}$

Step 3 (fast): $\text{N}_2\text{O} + \text{H}_2 \rightarrow \text{N}_2 + \text{H}_2\text{O}$

Show that this mechanism is consistent with your deduced rate equation.

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8 Arrhenius Calculation: Effect of Temperature Increase on Rate Constant — 9701/41/O/N/22/Q2 [6 Marks] HARD · Syllabus 26.4

The rate constant for a reaction is $1.50 \times 10^{-4} \text{ s}^{-1}$ at 300 K.

The activation energy for the reaction is 80.0 kJ mol^{-1} .

Gas constant $R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$.

The Arrhenius equation in two-temperature form is:

$$\ln(k_2 / k_1) = (-E_a / R) \times (1/T_2 - 1/T_1)$$

- (a) Calculate the value of the rate constant k_2 at 320 K. [3]

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- (b) Calculate the factor by which the rate of reaction increases when temperature is raised from 300 K to 320 K. [1]

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- (c) Using collision theory and the Maxwell-Boltzmann distribution, explain why a modest increase of 20 K causes a dramatic increase in reaction rate. [2]

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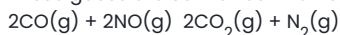
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9 Heterogeneous Catalysis: Catalytic Converter & Adsorption Desorption — 9701/42/M/J/21/Q3 [6 Marks] HARD · Syllabus 26.3

Automotive exhaust gases contain toxic pollutants including carbon monoxide (CO) and nitrogen monoxide (NO).

These gases are converted into non-toxic products across a catalytic converter containing platinum and rhodium catalysts:



- (a) Explain the meaning of the term heterogeneous catalyst in this reaction. [2]

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- (b) Describe the mode of action of the heterogeneous catalyst in terms of adsorption, bond weakening, reaction, and desorption. [3]

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- (c) Explain why leaded fuel permanently 'poisons' the catalytic converter. [1]

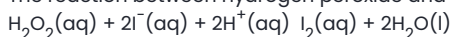
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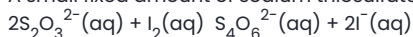
10 Iodine Clock Reaction: Experimental Design & Initial Rate Calculation — 9701/41/M/J/21/Q2 [6 Marks]

HARD · Syllabus 26.1

The reaction between hydrogen peroxide and iodide in acidic solution is used in the 'iodine clock' experiment:



A small fixed amount of sodium thiosulfate and starch indicator are added to each reaction mixture:



The time t for the mixture to suddenly turn blue-black is recorded.

- (a) Explain why the sudden appearance of the blue-black colour signals a specific amount of reaction. [2]

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- (b) Explain why $(1/t)$ can be taken as a direct measure of initial rate in this experiment. [2]

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- (c) State two precautions necessary to ensure that $(1/t)$ accurately represents the true initial rate. [2]

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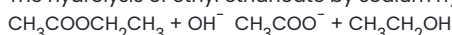
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11 Kinetics of Alkaline Hydrolysis of an Ester (Saponification) — 9701/42/O/N/21/Q3 [6 Marks]

HARD · Syllabus 26.1

The hydrolysis of ethyl ethanoate by sodium hydroxide was investigated:



The reaction is first order in ethyl ethanoate and first order in OH^- .

At 298 K, the rate constant $k = 0.110 \text{ mol}^{-1} \text{ dm}^3 \text{ s}^{-1}$.

- (a) Write the rate equation for this reaction and state the overall order. [2]

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- (b) Calculate the initial rate of reaction when 50.0 cm^3 of $0.200 \text{ mol dm}^{-3}$ ethyl ethanoate is mixed with 50.0 cm^3 of $0.100 \text{ mol dm}^{-3}$ NaOH. [2]

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- (c) Suggest how electrical conductivity can be used to continuously monitor the progress of this reaction. [2]

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12

Homogeneous Catalysis: Oxidation of Tartrate by Hydrogen Peroxide — 9701/41/O/N/21/Q2 [6 Marks]

HARD · Syllabus 26.3

The oxidation of potassium sodium tartrate by hydrogen peroxide is slow at 330 K:



When pink $\text{Co}^{2+}(\text{aq})$ ions are added, the mixture turns dark green with vigorous effervescence, before returning to its original pink colour once effervescence ceases.

- (a) Explain how these visual observations demonstrate that cobalt acts as a homogeneous catalyst. [2]

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- (b) Explain the role of transition metal ions in redox catalysis in terms of variable oxidation states. [2]

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- (c) Draw a sketch of a reaction energy profile comparing the catalysed and uncatalysed reaction pathways, labelling H and E_a . [2]

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13

Pseudo-First-Order Kinetics: Sucrose Inversion — 9701/42/M/J/20/Q2 [6 Marks]

HARD · Syllabus 26.1

The acid-catalysed hydrolysis of sucrose into glucose and fructose follows the rate law:

$$\text{Rate} = k[\text{sucrose}][\text{H}_2\text{O}][\text{H}^+]$$



Because water is the solvent, $[\text{H}_2\text{O}]$ is constant; 55.5 mol dm⁻³ and remains essentially constant.

$[\text{H}^+]$ is also constant as it is a catalyst.

- (a) Explain what is meant by a pseudo-first-order reaction in this context. [2]

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- (b) Write the pseudo-first-order rate equation and express the apparent rate constant k' in terms of k , $[\text{H}_2\text{O}]$, and $[\text{H}^+]$. [2]

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- (c) The apparent rate constant k' is $4.80 \times 10^{-4} \text{ s}^{-1}$. Calculate the time taken for 75% of the sucrose to be hydrolysed. [2]

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14

Arrhenius Parameters: Comparison of Catalysed vs Uncatalysed Decomposition of H₂O₂ — 9701/41/M/J/20/Q2 [6 Marks]

HARD · Syllabus 26.4

The decomposition of aqueous hydrogen peroxide, $2\text{H}_2\text{O}_2(\text{aq}) \rightarrow 2\text{H}_2\text{O}(\text{l}) + \text{O}_2(\text{g})$, has an uncatalysed activation energy of 75.0 kJ mol⁻¹.

In the presence of catalase enzyme, the activation energy drops to 8.00 kJ mol⁻¹.

Assume the frequency factor A is identical for both pathways at 298 K.

$$R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$$

- (a) State the Arrhenius equation in its exponential form, identifying all symbols. [2]

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- (b) Calculate the ratio of the catalysed rate constant to the uncatalysed rate constant, $k_{\text{cat}} / k_{\text{uncat}}$, at 298 K. [3]

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- (c) State why enzymes lose their catalytic activity at temperatures above 330 K (57 °C). [1]

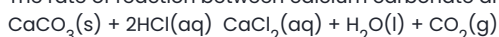
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15 Gas Collection Method & Initial Rate Deduction for $\text{CaCO}_3 + \text{HCl}$ — 9701/42/O/N/19/Q2 [6 Marks]

HARD · Syllabus 26.2

The rate of reaction between calcium carbonate and hydrochloric acid was investigated:



Carbon dioxide volume was collected in a gas syringe over time.

- (a) Describe how the initial rate of reaction is determined from a graph of volume of CO_2 against time. [2]

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- (b) Explain why small marble chips react faster than an equal mass of large marble chips of the same chemical purity. [2]

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- (c) Explain why the rate of reaction eventually falls to zero, and state which reagent was in excess if some solid carbonate remains. [2]

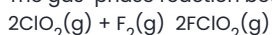
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16 Deduction of Reaction Mechanism from Complex Rate Law — 9701/41/O/N/19/Q2 [6 Marks]

HARD · Syllabus 26.3

The gas-phase reaction between chlorine dioxide and fluorine was investigated:



The experimentally determined rate equation is: $\text{Rate} = k[\text{ClO}_2][\text{F}_2]$.

- (a) Explain why the reaction cannot take place in a single termolecular collision step involving 2 molecules of ClO_2 and 1 molecule of F_2 . [2]

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- (b) Propose a two-step reaction mechanism consistent with the observed rate equation, identifying which step is rate-determining. [2]

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- (c) Identify any intermediate species in your proposed mechanism and explain why it does not appear in the overall stoichiometric equation. [2]

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SECTION B: 4-MARK STRUCTURED EXAM QUESTIONS (40% TARIFF · Q17–Q32)

17 Orders of Reaction Definitions & Meaning of Overall Order — 9701/42/M/J/23/Q5 [4 Marks]

MEDIUM · Syllabus 26.1

The rate equation for a reaction is given by: $\text{Rate} = k[\text{A}]^m[\text{B}]^n$.

(a) Define the term order of reaction with respect to reactant A. [2]

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(b) Define the overall order of reaction, and state its value if $m = 1$ and $n = 2$. [2]

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18 Units of Rate Constant k for Different Orders — 9701/41/M/J/23/Q5 [4 Marks]

MEDIUM · Syllabus 26.1

The units of the rate constant k vary with the overall order of the reaction.

(a) Derive the units of k for a second-order reaction: $\text{Rate} = k[\text{X}]^2$. [2]

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(b) Derive the units of k for a third-order reaction: $\text{Rate} = k[\text{X}][\text{Y}]^2$. [2]

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19 Half-Life and First-Order Rate Constant Calculation — 9701/42/O/N/23/Q5 [4 Marks]

MEDIUM · Syllabus 26.2

Cyclopropane isomerises into propene in a first-order gas-phase reaction.

At 773 K, the rate constant $k = 3.30 \times 10^{-4} \text{ s}^{-1}$.

(a) Calculate the half-life, $t_{1/2}$, of cyclopropane in seconds. [2]

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(b) Calculate the fraction of cyclopropane remaining after 4200 seconds. [2]

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20 Rate-Determining Step in $\text{S}_{\text{N}}2$ Mechanism — 9701/41/O/N/23/Q5 [4 Marks]

MEDIUM · Syllabus 26.3

Bromoethane reacts with aqueous hydroxide ions by an $\text{S}_{\text{N}}2$ mechanism:



$\text{Rate} = k[\text{CH}_3\text{CH}_2\text{Br}][\text{OH}^-]$.

(a) Explain what the term molecularity means for this elementary step. [2]

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- (b) Draw the five-coordinate transition state formed during this single-step reaction. [2]

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21 Quenching and Sampling Method in Ester Hydrolysis — 9701/42/M/J/22/Q4 [4 Marks]

MEDIUM · Syllabus 26.2

The rate of hydrolysis of ethyl ethanoate by acid was monitored by sampling and titration:



- (a) Explain what is meant by quenching a reaction mixture. [2]

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- (b) Suggest two ways this reaction mixture can be quenched before titration. [2]

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22 Effect of Catalyst on Activation Energy and Equilibrium — 9701/41/M/J/22/Q4 [4 Marks]

MEDIUM · Syllabus 26.3

The Haber process reaction: $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$ $\Delta H = -92 \text{ kJ mol}^{-1}$ uses a finely divided iron catalyst.

- (a) Explain how the iron catalyst increases the rate of reaction. [2]

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- (b) Explain the effect, if any, of the iron catalyst on the yield of ammonia at equilibrium. [2]

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23 Arrhenius Straight-Line Equation Transformation — 9701/42/O/N/22/Q4 [4 Marks]

MEDIUM · Syllabus 26.4

The Arrhenius equation is $k = A \exp(-E_a / RT)$.

- (a) Take natural logarithms of both sides to convert this equation into the linear form $y = mx + c$. [2]

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- (b) Identify the variables plotted on the y-axis and x-axis, and state what the gradient and y-intercept represent. [2]

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24 Continuous Monitoring by Electrical Conductivity — 9701/41/O/N/22/Q4 [4 Marks]

MEDIUM · Syllabus 26.2

Consider the reaction: $(\text{CH}_3)_3\text{C-Cl} + \text{H}_2\text{O} \rightarrow (\text{CH}_3)_3\text{C-OH} + \text{H}^+(\text{aq}) + \text{Cl}^-(\text{aq})$.

- (a) Explain why electrical conductivity is an ideal method for continuously monitoring the rate of this reaction. [2]

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- (b) Describe how the conductivity changes over time, and state how initial rate is determined from the graph. [2]
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25 Enzyme Kinetics & Substrate Concentration Effect — 9701/42/M/J/21/Q4 [4 Marks]

MEDIUM · Syllabus 26.3

The hydrolysis of urea by the enzyme urease was studied at varying substrate concentrations.

- (a) State the order of reaction with respect to urea at very low substrate concentration, and explain why. [2]
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- (b) State the order of reaction with respect to urea at very high substrate concentration, and explain why. [2]
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26 Half-Life Comparison: First Order vs Second Order Kinetics — 9701/41/M/J/21/Q4 [4 Marks]

MEDIUM · Syllabus 26.2

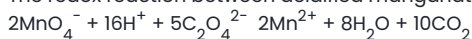
Reaction 1 is first order in reactant A. Reaction 2 is second order in reactant B.
Both have an initial half-life of 60 seconds with initial concentration 1.00 mol dm^{-3} .

- (a) State the value of the second half-life for Reaction 1, giving your reasoning. [2]
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- (b) State what happens to the half-life of Reaction 2 as the reaction proceeds, giving your reasoning. [2]
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27 Autocatalysis: Reaction Between Permanganate and Ethanedioate — 9701/42/O/N/21/Q4 [4 Marks]

MEDIUM · Syllabus 26.3

The redox reaction between acidified manganate(VII) and ethanedioic acid is initially very slow:



After a short induction period, the purple colour fades rapidly.

- (a) Explain why the reaction is initially very slow at room temperature. [2]
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- (b) Explain why the rate of reaction accelerates dramatically after a short time, identifying the autocatalyst. [2]
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28 Calculating Activation Energy from Two Rate Constants — 9701/41/O/N/21/Q4 [4 Marks]

MEDIUM · Syllabus 26.4

For a certain chemical reaction, $k = 2.00 \times 10^{-3} \text{ s}^{-1}$ at 293 K and $k = 8.00 \times 10^{-3} \text{ s}^{-1}$ at 313 K.
 $R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$.

- (a) Calculate the value of $\ln(k_2 / k_1)$ and $(1/T_2 - 1/T_1)$. [2]
-
-

- (b) Calculate the activation energy E_a in kJ mol^{-1} . [2]

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29 Deducing Rate Equation from Initial Rates Table — 9701/42/M/J/20/Q4 [4 Marks]

MEDIUM · Syllabus 26.1

Data for $2A + B \rightarrow C$:

- Expt 1: $[A] = 0.10$, $[B] = 0.10$, Rate = $1.2 \times 10^{-3} \text{ mol dm}^{-3} \text{ s}^{-1}$
- Expt 2: $[A] = 0.20$, $[B] = 0.10$, Rate = $2.4 \times 10^{-3} \text{ mol dm}^{-3} \text{ s}^{-1}$
- Expt 3: $[A] = 0.20$, $[B] = 0.20$, Rate = $9.6 \times 10^{-3} \text{ mol dm}^{-3} \text{ s}^{-1}$

- (a) Deduce the order with respect to A and with respect to B. [2]

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- (b) State the rate equation and calculate k. [2]

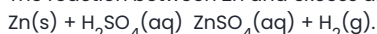
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30 Mass Loss Method for Solid-Liquid Reaction Kinetics — 9701/41/M/J/20/Q5 [4 Marks]

MEDIUM · Syllabus 26.2

The reaction between Zn and excess dilute H_2SO_4 liberates H_2 gas:



- (a) Explain why measuring mass loss on a balance is suitable for this reaction, and name a piece of apparatus placed in the flask neck. [2]

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- (b) Explain how the rate at time $t = 60 \text{ s}$ is calculated from the mass loss curve. [2]

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31 Molecularity and Intermediate in Multi-Step Mechanism — 9701/42/O/N/19/Q4 [4 Marks]

MEDIUM · Syllabus 26.3

A reaction proceeds by the following mechanism:

Step 1: $\text{NO}_2 + \text{NO}_2 \rightarrow \text{NO}_3 + \text{NO}$ (slow)

Step 2: $\text{NO}_3 + \text{CO} \rightarrow \text{NO}_2 + \text{CO}_2$ (fast)

- (a) Write the overall balanced equation for the reaction. [2]

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- (b) Deduce the rate equation and identify the intermediate species. [2]

.....

.....

32 Activation Energy and Pre-Exponential Factor Concept — 9701/41/O/N/19/Q4 [4 Marks]

MEDIUM · Syllabus 26.4

The Arrhenius parameters for two reactions X and Y are:

- Reaction X: $E_a = 50 \text{ kJ mol}^{-1}$, $A = 1.0 \times 10^{11} \text{ s}^{-1}$
- Reaction Y: $E_a = 90 \text{ kJ mol}^{-1}$, $A = 1.0 \times 10^{14} \text{ s}^{-1}$

- (a) State which reaction has a faster rate at 298 K, giving your reasoning. [2]
-
-
- (b) Explain which reaction's rate will show a greater sensitivity to an increase in temperature. [2]
-
-

SECTION C: 2-MARK TARGETED EXAM QUESTIONS (20% TARIFF · Q33–Q40)

33 Definition of Rate of Reaction — 9701/42/M/J/23/Q6 [2 Marks]

EASY · Syllabus 26.1

Chemical kinetics studies the speed of chemical reactions.

- (a) Define the term rate of reaction. [2]
-
-

34 Definition of Half-Life — 9701/41/M/J/23/Q6 [2 Marks]

EASY · Syllabus 26.2

The half-life is a fundamental kinetic property of reactions.

- (a) Define the term half-life, $t_{1/2}$. [2]
-
-

35 Definition of Rate-Determining Step — 9701/42/O/N/23/Q6 [2 Marks]

EASY · Syllabus 26.3

Many chemical reactions occur via several individual stages.

- (a) Define what is meant by the rate-determining step of a reaction. [2]
-
-

36 Definition of Activation Energy — 9701/41/O/N/23/Q6 [2 Marks]

EASY · Syllabus 26.4

Chemical collisions must satisfy an energy threshold.

- (a) Define the term activation energy, E_a . [2]
-
-

37 Definition of a Catalyst — 9701/42/M/J/22/Q5 [2 Marks]

EASY · Syllabus 26.3

Catalysts are widely used in chemical manufacture.

- (a) Define the term catalyst. [2]
-
-

38 Homogeneous versus Heterogeneous Catalysis Distinction — 9701/41/M/J/22/Q5 [2 Marks]

EASY · Syllabus 26.3

Catalysts operate either in the same phase or in different phases.

- (a) Distinguish between homogeneous and heterogeneous catalysis.

[2]

39 Pre-Exponential Factor A Meaning — 9701/42/O/N/22/Q5 [2 Marks]

EASY · Syllabus 26.4

In the Arrhenius equation $k = A \exp(-E_a / RT)$, A is a constant.

- (a) State what the pre-exponential factor A represents.

[2]

40 Rate Constant Independence of Concentration — 9701/41/O/N/22/Q5 [2 Marks]

EASY · Syllabus 26.1

A student changes the concentration of reactants in an experiment at constant temperature.

- (a) State and explain whether the rate constant k changes when reactant concentrations are tripled at constant temperature.

[2]

SECTION D: HIGH-FREQUENCY CORE REPEATS — 10 MOST FREQUENTLY TESTED QUESTIONS (Q41–Q50)

41 [CORE REPEAT 1] Complete Arrhenius Graphical Analysis — 9701/42/M/J/23/Q3 [6 Marks]

HARD · Syllabus 26.4

The decomposition of benzenediazonium chloride, $C_6H_5N_2Cl(aq) \rightarrow C_6H_5Cl(aq) + N_2(g)$, was monitored at various temperatures. An Arrhenius plot of $\ln k$ against $1/T$ yields a linear plot identical to Fig. 41.1.

Gradient = $-1.25 \times 10^4 \text{ K}$.

$R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$.

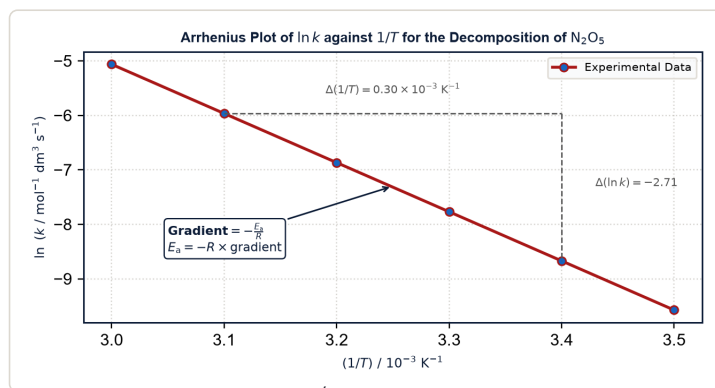


Fig. 41.1: Arrhenius plot of $\ln k$ against $1/T$ for benzenediazonium chloride decomposition.

- (a) Calculate the activation energy E_a for this reaction in kJ mol^{-1} .

[2]

- (b) At 320 K, $k = 1.40 \times 10^{-3} \text{ s}^{-1}$. Use this value and your E_a to calculate the pre-exponential factor A. [2]
- (c) State what the units of A are for this reaction, giving your reasoning. [2]

42 [CORE REPEAT 2] Initial Rates Deductions & Rate Law Construction — 9701/41/M/J/23/Q3 [6 Marks]

HARD · Syllabus 26.1

Reaction: $2\text{NO}(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{NO}_2(\text{g})$.

- Expt 1: $[\text{NO}] = 0.010$, $[\text{O}_2] = 0.010$, Rate = $7.00 \times 10^{-4} \text{ mol dm}^{-3} \text{ s}^{-1}$
- Expt 2: $[\text{NO}] = 0.010$, $[\text{O}_2] = 0.020$, Rate = $1.40 \times 10^{-3} \text{ mol dm}^{-3} \text{ s}^{-1}$
- Expt 3: $[\text{NO}] = 0.030$, $[\text{O}_2] = 0.020$, Rate = $1.26 \times 10^{-2} \text{ mol dm}^{-3} \text{ s}^{-1}$

- (a) Deduce the order of reaction with respect to NO and with respect to O_2 . [2]
- (b) Write the rate equation and calculate the rate constant k with its units. [2]
- (c) Calculate the rate of reaction when $[\text{NO}] = 0.025 \text{ mol dm}^{-3}$ and $[\text{O}_2] = 0.015 \text{ mol dm}^{-3}$. [2]

43 [CORE REPEAT 3] Constant Half-Life Verification & S_N1 Reaction Pathway — 9701/42/O/N/23/Q3 [6 Marks]

HARD · Syllabus 26.2

The decay curve for reactant R is shown in Fig. 43.1.

The initial concentration is 0.80 mol dm^{-3} .

Concentrations at successive time intervals:

- $t = 0 \text{ s}$, $[\text{R}] = 0.80 \text{ mol dm}^{-3}$
- $t = 40 \text{ s}$, $[\text{R}] = 0.40 \text{ mol dm}^{-3}$
- $t = 80 \text{ s}$, $[\text{R}] = 0.20 \text{ mol dm}^{-3}$
- $t = 120 \text{ s}$, $[\text{R}] = 0.10 \text{ mol dm}^{-3}$

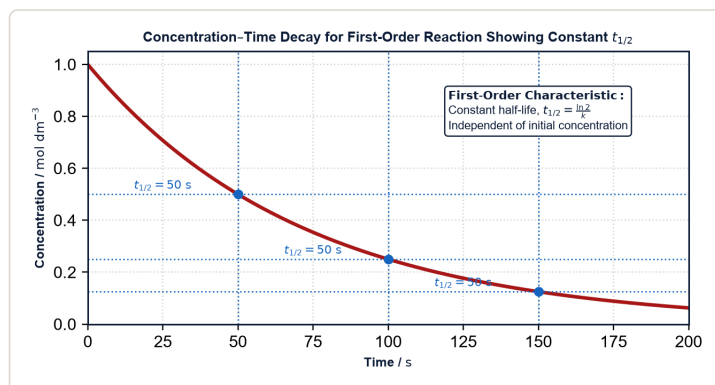


Fig. 43.1: Decay curve demonstrating constant successive half-lives of 40 s.

- (a) Prove that the reaction is first order by analysing the three successive half-lives. [2]
- (b) Calculate the rate constant k for this first-order decomposition. [2]
- (c) If R is 2-chloro-2-methylpropane undergoing substitution with OH^- , explain why the rate is independent of $[\text{OH}^-]$. [2]

44 [CORE REPEAT 4] Two-Step Mechanism Verification & Intermediate Identification — 9701/41/O/N/23/Q3 [6 Marks] Syllabus 26.3

Reaction: $2\text{NO}_2(\text{g}) + \text{F}_2(\text{g}) \rightarrow 2\text{NO}_2\text{F}(\text{g})$.

The reaction coordinate diagram in Fig. 44.1 shows two transition states with $E_{a1} > E_{a2}$.

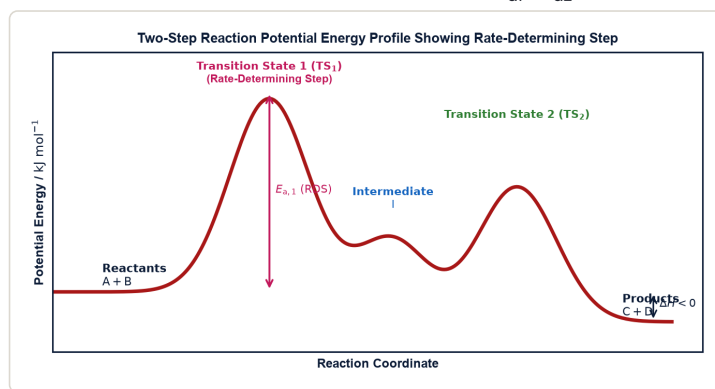


Fig. 44.1: Energy profile showing two transition states and intermediate F atom.

- (a) Explain why Step 1 ($\text{NO}_2 + \text{F}_2 \rightarrow \text{NO}_2\text{F} + \text{F}$) is the rate-determining step. [2]
- (b) Deduce the rate equation consistent with this two-step mechanism. [2]
- (c) State the role of the fluorine atom F, and explain why it is not classified as a catalyst. [2]

45 [CORE REPEAT 5] Zero-Order Kinetics of Iodine Reaction — 9701/42/M/J/22/Q2 [4 Marks]

MEDIUM · Syllabus 26.2

In the reaction $\text{CH}_3\text{COCH}_3 + \text{I}_2 + \text{H}^+ \rightarrow \text{products}$, the order with respect to I_2 is zero.

- (a) Sketch the concentration of I_2 versus time curve, labelling axes clearly. [2]

- (b) Explain why the rate of reaction remains unchanged even when $[I_2]$ is increased five-fold. [2]

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46 [CORE REPEAT 6] Rate Constant Calculation from Rate and Concentrations — 9701/41/M/J/22/Q2 [4 Marks] MEDIUM · Syllabus 26.1

For the reaction $A + 2B \rightarrow C$, Rate = $k[A][B]$.

When $[A] = 0.0500 \text{ mol dm}^{-3}$ and $[B] = 0.0800 \text{ mol dm}^{-3}$, the rate of reaction is $3.60 \times 10^{-4} \text{ mol dm}^{-3} \text{ s}^{-1}$.

- (a) Calculate the value of the rate constant k . [2]

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- (b) State the units of k and explain why the stoichiometry of $2B$ does not appear as an exponent in the rate equation. [2]

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47 [CORE REPEAT 7] Two-Point Arrhenius Calculation of E_a — 9701/42/O/N/22/Q2 [4 Marks] MEDIUM · Syllabus 26.4

A reaction has rate constant $k_1 = 2.50 \times 10^{-4} \text{ s}^{-1}$ at 300 K and $k_2 = 1.00 \times 10^{-3} \text{ s}^{-1}$ at 320 K.

$R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$.

- (a) Calculate $\ln(k_2 / k_1)$. [1]

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- (b) Calculate the activation energy E_a in kJ mol^{-1} . [3]

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48 [CORE REPEAT 8] Homogeneous Catalysis Mechanism of Iodide-Persulfate — 9701/41/O/N/22/Q2 [4 Marks] MEDIUM · Syllabus 26.3

$\text{Fe}^{2+}(\text{aq})$ can also act as an efficient catalyst for the reaction between $\text{S}_2\text{O}_8^{2-}$ and I^- .

Standard electrode potentials:

$\text{S}_2\text{O}_8^{2-} + 2\text{e}^- \rightarrow 2\text{SO}_4^{2-}$, $E^\circ = +2.01 \text{ V}$

$\text{Fe}^{3+} + \text{e}^- \rightarrow \text{Fe}^{2+}$, $E^\circ = +0.77 \text{ V}$

$\text{I}_2 + 2\text{e}^- \rightarrow 2\text{I}^-$, $E^\circ = +0.54 \text{ V}$

- (a) Write two equations showing how Fe^{2+} catalyses this reaction. [2]

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- (b) Explain why both steps are energetically feasible by quoting E° values. [2]

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49 [CORE REPEAT 9] First-Order Rate Equation Expression & Definition — 9701/42/M/J/21/Q2 [2 Marks] EASY · Syllabus 26.1

A reaction $A \rightarrow \text{products}$ is first order.

- (a) Write the differential rate equation and state the relation between $t_{1/2}$ and k . [2]

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[CORE REPEAT 10] Transition State vs Intermediate Difference — 9701/41/M/J/21/Q2 [2 Marks]

EASY · Syllabus 26.3

Reaction profiles distinguish between transition states and intermediates.

- (a) State one structural and one energetic difference between an intermediate and a transition state. [2]

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Mark Scheme — Topic 26 — Reaction Kinetics

CAMBRIDGE INTERNATIONAL A LEVEL CHEMISTRY (9701) — PAPER 4 MARKING CRITERIA

Q	EXAMINER MARKING POINTS / WORKED MATHEMATICAL DERIVATIONS	MARKS
1(a)	Gradient = $y / x = (-8.67 - (-5.97)) / ((3.40 - 3.10) \times 10^{-3}) = -2.70 / (0.30 \times 10^{-3})$ [1]; $= -9.00 \times 10^3 \text{ K (or } -9000 \text{ K)}$ [1].	2
1(b)	Gradient = $-E_a / R$ &RArr; $E_a = -R \times \text{gradient} = -8.314 \times (-9000) = +74826 \text{ J mol}^{-1}$ [1]; $E_a = +74.8 \text{ kJ mol}^{-1}$ (allow 74.5 to 75.2 kJ mol ⁻¹) [1].	2
1(c)	$\ln k = \ln A - (E_a / RT)$ &RArr; $-5.97 = \ln A - (9000 \times 3.10 \times 10^{-3}) = \ln A - 27.90$ &RArr; $\ln A = 21.93$ &RArr; $A = e^{21.93} = 3.34 \times 10^9 \text{ s}^{-1}$ [1]; A represents the frequency of collisions with correct orientation (collision frequency factor) [1].	2
2(a)	Comparing Expt 1 and 2: [S2O8 2-] doubles while [I-] is constant; Rate doubles (4.80/2.40 = 2), so order with respect to S2O8 2- is 1 [1]; Comparing Expt 2 and 3: [I-] triples while [S2O8 2-] is constant; Rate triples (1.44/0.48 = 3), so order with respect to I- is 1 [1]; Overall order = 1 + 1 = 2 [1].	3
2(b)	Rate = $k[\text{S2O8 2-}][\text{I-}]$ [1]; $k = \text{Rate} / ([\text{S2O8 2-}][\text{I-}]) = (2.40 \times 10^{-5}) / (0.040 \times 0.020) = 0.030 \text{ mol}^{-1} \text{ dm}^3 \text{ s}^{-1}$ [1].	2
2(c)	Step 1: $2\text{Fe}^{3+} + 2\text{I}^- \rightarrow 2\text{Fe}^{2+} + \text{I}_2$; Step 2: $2\text{Fe}^{2+} + \text{S2O8 2-} \rightarrow 2\text{Fe}^{3+} + 2\text{SO4 2-}$ [1].	1
3(a)	For a first-order reaction, the half-life $t_{1/2} = \ln 2 / k$, which is independent of the initial or instantaneous reactant concentration [1]; The successive half-lives remain constant at approximately 48 s, confirming first-order kinetics [1].	2
3(b)	$k = \ln 2 / t_{1/2} = 0.693 / 48.3 \text{ s}$ [1]; $k = 0.0143 \text{ s}^{-1}$ (units: s ⁻¹) [1].	2
3(c)	Rate = $k[(\text{CH}_3)_3\text{CBr}]$ [1]; Step 1 (slow/RDS): $(\text{CH}_3)_3\text{CBr} \rightarrow (\text{CH}_3)_3\text{C}^+ + \text{Br}^-$; Step 2 (fast): $(\text{CH}_3)_3\text{C}^+ + \text{OH}^- \rightarrow (\text{CH}_3)_3\text{COH}$ [1].	2
4(a)	Step 1 is the rate-determining step [1]; It possesses a much higher activation energy barrier ($E_{a1} = 65 \text{ kJ mol}^{-1}$ vs $E_{a2} = 20 \text{ kJ mol}^{-1}$), making it the slowest elementary step [1].	2
4(b)	Since the slow step involves one molecule of A and one molecule of B [1]; Rate = $k[A][B]$ [1].	2
4(c)	An intermediate represents a real chemical species located at a local potential energy minimum with fully formed bonds that has a finite lifetime [1]; A transition state represents a high-energy maximum corresponding to partially formed and broken bonds and cannot be isolated [1].	2
5(a)	The graph of absorbance (proportional to [I2]) against time is a straight line with a constant negative slope [1]; This shows that the rate of disappearance of I2 is constant and independent of [I2], confirming zero order [1].	2
5(b)	Rate = $-(\text{d}[I_2] / \text{d}t) = -(\text{gradient of absorbance curve} / \text{calibration factor})$ [1]; Constant gradient directly gives the constant rate of reaction [1].	2
5(c)	H ⁺ acts as a homogeneous catalyst and is involved in the rate-determining step before being regenerated later [1]; Protonation of the carbonyl oxygen of propanone in the slow enolisation step: $\text{CH}_3\text{COCH}_3 + \text{H}^+ \rightarrow \text{CH}_3\text{C}(\text{OH}^+)\text{CH}_3$ [1].	2
6(a)	Graph 1 (horizontal): Zero order (rate is independent of [A]) [1]; Graph 2 (straight line through origin): First order (rate is directly proportional to [A]) [1]; Graph 3 (upward parabola): Second order (rate is proportional to [A] ²) [1].	3
6(b)	Zero order: $k = \text{Rate} / [A]^0$ &RArr; $\text{mol dm}^{-3} \text{ s}^{-1}$ [1]; First order: $k = \text{Rate} / [A]^1$ &RArr; s^{-1} [1]; Second order: $k = \text{Rate} / [A]^2$ &RArr; $\text{mol}^{-1} \text{ dm}^3 \text{ s}^{-1}$ [1].	3
7(a)	Expt 1 to 2: [NO] doubles, rate quadruples (1.20/0.30 = 4 = 2 ²) &RArr; 2nd order with respect to NO [1]; Expt 2 to 3: [H2] doubles, rate doubles (2.40/1.20 = 2) &RArr; 1st order with respect to H2 [1].	2
7(b)	Rate = $k[\text{NO}]^2[\text{H}_2]$ [1]; $k = (3.00 \times 10^{-5}) / ((0.0020)^2 \times 0.0010) = 7.50 \times 10^3 \text{ mol}^{-2} \text{ dm}^6 \text{ s}^{-1}$ [1].	2
7(c)	Step 2 is the RDS: Rate = $k_2[\text{N}_2\text{O}_2][\text{H}_2]$ [1]; From fast pre-equilibrium Step 1: $K_{eq} = [\text{N}_2\text{O}_2] / [\text{NO}]^2$ &RArr; $[\text{N}_2\text{O}_2] = K_{eq}[\text{NO}]^2$; Substituting gives Rate = $k_2 K_{eq} [\text{NO}]^2[\text{H}_2] = k[\text{NO}]^2[\text{H}_2]$, matching experimental rate equation [1].	2
8(a)	$1/T_2 - 1/T_1 = 1/320 - 1/300 = 0.003125 - 0.003333 = -0.000208 \text{ K}^{-1}$ [1]; $\ln(k_2/k_1) = (-80000 / 8.314) \times (-0.000208) = +2.004$ [1]; $k_2/k_1 = e^{2.004} = 7.42$ &RArr; $k_2 = 7.42 \times (1.50 \times 10^{-4}) = 1.11 \times 10^{-3} \text{ s}^{-1}$ [1].	3
8(b)	Factor of increase = $k_2 / k_1 = 7.42$ (rate increases by approximately 7.4 times) [1].	1
8(c)	Higher temperature increases the mean kinetic energy of particles, shifting the Maxwell-Boltzmann distribution to the right [1]; A significantly greater proportion/fraction of colliding molecules possess energy greater than or equal to the activation energy ($E \geq E_a$) [1].	2

Q	EXAMINER MARKING POINTS / WORKED MATHEMATICAL DERIVATIONS	MARKS
9(a)	A catalyst that exists in a different physical phase/state from the reactants [1]; The catalyst is solid (Pt/Rh) while reactants CO and NO are gases [1].	2
9(b)	1. Adsorption: Reactant gas molecules form weak bonds to active sites on the metal surface [1]; 2. Reaction: Covalent bonds within reactant molecules are weakened, lowering E_a , and molecules are held in favourable orientation to react together [1]; 3. Desorption: Product molecules (CO ₂ and N ₂) break bonds with surface and detach, freeing active sites for further reactants [1].	3
9(c)	Lead atoms adsorb strongly and irreversibly onto the active sites of the metal surface, blocking reactant molecules from binding [1].	1
10(a)	Thiosulfate rapidly reacts with and consumes liberated I ₂ as fast as it forms [1]; Once all S ₂ O ₃ ²⁻ has been completely exhausted, excess free I ₂ instantly complexes with starch to form the blue-black species [1].	2
10(b)	Initial rate = $[I_2] / t$; Because the moles of S ₂ O ₃ ²⁻ added is kept constant across all runs, $[I_2]$ is constant [1]; Therefore, initial rate is directly proportional to $1/t$ [1].	2
10(c)	Keep temperature strictly constant (using a thermostatted water bath) [1]; Ensure the amount of S ₂ O ₃ ²⁻ added is very small so that less than 15% of reactants are consumed during the timing period [1].	2
11(a)	Rate = $k[CH_3COOCH_2CH_3][OH^-]$ [1]; Overall order = $1 + 1 = 2$ (second order) [1].	2
11(b)	Upon mixing equal volumes, concentrations are halved: $[ester] = 0.100 \text{ mol dm}^{-3}$, $[OH^-] = 0.0500 \text{ mol dm}^{-3}$ [1]; Rate = $0.110 \times (0.100) \times (0.0500) = 5.50 \times 10^{-4} \text{ mol dm}^{-3} \text{ s}^{-1}$ [1].	2
11(c)	As the reaction proceeds, highly mobile OH ⁻ ions (high molar ionic conductivity) are replaced by much bulkier, slower CH ₃ COO ⁻ ions [1]; The electrical conductivity of the solution steadily decreases over time; the rate of decrease is directly proportional to reaction rate [1].	2
12(a)	Co ²⁺ is in the same aqueous phase as reactants (homogeneous) [1]; Pink Co ²⁺ is oxidised to a green cobalt(III)-tartrate intermediate during reaction and then reformed as pink Co ²⁺ at the end unchanged in mass and chemical identity [1].	2
12(b)	Transition metals can readily interchange between stable oxidation states (e.g. Co ²⁺ and Co ³⁺) [1]; This provides an alternative two-step pathway with lower activation energy for electron transfer [1].	2
12(c)	Profile showing single high peak for uncatalysed and two lower peaks for catalysed pathway [1]; Correct labelling of H (same for both) and $E_a(uncat) > E_a(cat)$ [1].	2
13(a)	A multi-order reaction that behaves experimentally as first-order [1]; because all other reacting species are present in large excess (or constant concentration) so their concentrations do not change appreciably [1].	2
13(b)	Rate = $k'[\text{sucrose}]$ [1]; where $k' = k[H_2O][H^+]$ [1].	2
13(c)	75% hydrolysed means 25% remains &Arr; two successive half-lives (100% 50% 25%) [1]; $t_{1/2} = \ln 2 / k' = 0.693 / (4.80 \times 10^{-4}) = 1444 \text{ s}$ &Arr; Total time = $2 \times 1444 = 2888 \text{ s}$ (48.1 min) [1].	2
14(a)	$k = A e^{(-E_a / RT)}$ [1]; k = rate constant, A = pre-exponential factor, E_a = activation energy, R = gas constant, T = absolute temperature [1].	2
14(b)	$k_{cat} / k_{uncat} = e^{(-E_a(cat)/RT)} / e^{(-E_a(uncat)/RT)} = e^{((E_a(uncat) - E_a(cat)) / RT)}$ [1]; $E_a = 75000 - 8000 = 67000 \text{ J mol}^{-1}$; $RT = 8.314 \times 298 = 2477.6 \text{ J mol}^{-1}$ [1]; Ratio = $e^{(67000 / 2477.6)} = e^{27.042} = 5.55 \times 10^{11}$ (rate increases by over 500 billion times) [1].	3
14(c)	Thermal denaturation: hydrogen bonds and ionic interactions maintaining tertiary protein structure are disrupted, changing the shape of the active site [1].	1
15(a)	Draw a tangent to the curve at time $t = 0$ [1]; Calculate the gradient of this tangent (v / t) [1].	2
15(b)	Smaller chips have a much larger total surface area per unit mass [1]; More reactant particles are exposed, leading to a higher frequency of successful collisions between CaCO ₃ and H ⁺ ions [1].	2
15(c)	The limiting reactant (HCl) is completely consumed so $[H^+]$ falls to zero [1]; If unreacted CaCO ₃ remains, CaCO ₃ is in excess and HCl was the limiting reagent [1].	2
16(a)	A simultaneous collision between three gaseous molecules (termolecular) with correct orientation and sufficient energy is statistically extremely improbable [1]; If it were single-step, the rate law would be Rate = $k[ClO_2]^2[F_2]$, which contradicts experiment [1].	2

Q	EXAMINER MARKING POINTS / WORKED MATHEMATICAL DERIVATIONS	MARKS
16(b)	Step 1 (slow/RDS): $\text{ClO}_2 + \text{F}_2 \rightarrow \text{FClO}_2 + \text{F}$ [1]; Step 2 (fast): $\text{ClO}_2 + \text{F} \rightarrow \text{FClO}_2$ [1].	2
16(c)	Fluorine radical (or atom) F is the intermediate [1]; It is produced in Step 1 and subsequently consumed completely in Step 2, so it cancels out of the overall balanced equation [1].	2
17(a)	The power to which the concentration of reactant A is raised in the experimental rate equation [2].	2
17(b)	The sum of the powers of all concentration terms in the rate equation ($m + n$) [1]; Overall order = $1 + 2 = 3$ (third order) [1].	2
18(a)	$k = \text{Rate} / [\text{X}]^2 = (\text{mol dm}^{-3} \text{ s}^{-1}) / (\text{mol dm}^{-3})^2$ [1]; Units = $\text{mol}^{-1} \text{ dm}^3 \text{ s}^{-1}$ [1].	2
18(b)	$k = \text{Rate} / ([\text{X}][\text{Y}]^2) = (\text{mol dm}^{-3} \text{ s}^{-1}) / (\text{mol dm}^{-3})^3$ [1]; Units = $\text{mol}^{-2} \text{ dm}^6 \text{ s}^{-1}$ [1].	2
19(a)	$t_{1/2} = \ln 2 / k = 0.69315 / (3.30 \times 10^{-4} \text{ s}^{-1})$ [1]; $t_{1/2} = 2.10 \times 10^3 \text{ s}$ (2100 s) [1].	2
19(b)	Number of half-lives elapsed = $4200 / 2100 = 2$ half-lives [1]; Fraction remaining = $(1/2)^2 = 1/4 = 0.25$ (25%) [1].	2
20(a)	Molecularity is the number of reactant species colliding/involved in an elementary step [1]; It is bimolecular because one molecule of bromoethane and one hydroxide ion collide simultaneously [1].	2
20(b)	Central carbon with planar H, H, CH ₃ bonds, and partial bonds to incoming HO δ^- and outgoing δ^- -Br with brackets and negative charge [2].	2
21(a)	Stopping or dramatically slowing down the reaction at an exact recorded time so that reactant/product concentrations do not change before analysis [2].	2
21(b)	1. Rapid cooling by plunging the sample into an ice-water bath [1]; 2. Massive dilution with large volume of ice-cold water [1].	2
22(a)	Provides an alternative reaction pathway with lower activation energy (E_a) [1]; A greater fraction of collisions possess energy exceeding this lower E_a [1].	2
22(b)	No effect on the position of equilibrium or yield of ammonia [1]; A catalyst speeds up the rate of forward and reverse reactions by the exact same factor without changing K_c [1].	2
23(a)	$\ln k = (-E_a / R)(1/T) + \ln A$ [2].	2
23(b)	y-axis = $\ln k$; x-axis = $1/T$ [1]; Gradient = $-E_a / R$; y-intercept = $\ln A$ [1].	2
24(a)	The reactants are non-ionic (zero conductivity), whereas the reaction generates ions (H^+ and Cl^-) in solution [2].	2
24(b)	Conductivity increases steadily from zero as ion concentration rises [1]; The initial rate is found by measuring the initial gradient of the conductivity versus time curve [1].	2
25(a)	First order [1]; At low [urea], many enzyme active sites are unoccupied; rate is directly proportional to substrate concentration [1].	2
25(b)	Zero order [1]; At high [urea], all enzyme active sites are fully saturated (V_{max} reached); adding more substrate cannot increase rate [1].	2
26(a)	60 seconds (constant) [1]; First-order half-life is independent of concentration ($t_{1/2} = \ln 2 / k$) [1].	2
26(b)	The second half-life doubles to 120 seconds [1]; For a second-order reaction, $t_{1/2} = 1 / (k[\text{B}]_0)$, so half-life is inversely proportional to concentration; as $[\text{B}]$ halves, $t_{1/2}$ doubles [1].	2
27(a)	Both reactant ions (MnO_4^- and $\text{C}_2\text{O}_4^{2-}$) are negatively charged [1]; Strong electrostatic repulsion between like-charged anions results in a high activation energy [1].	2
27(b)	Autocatalysis: one of the products formed, $\text{Mn}^{2+}(\text{aq})$, acts as a catalyst [1]; Mn^{2+} reacts with MnO_4^- to form intermediate manganese species (Mn^{3+}) which rapidly oxidise ethanedioate [1].	2
28(a)	$\ln(k_2/k_1) = \ln(8.00/2.00) = \ln(4.00) = 1.386$ [1]; $1/313 - 1/293 = 0.003195 - 0.003413 = -2.180 \times 10^{-4} \text{ K}^{-1}$ [1].	2
28(b)	$E_a = -R \times \ln(k_2/k_1) / (1/T_2 - 1/T_1) = -8.314 \times (1.386) / (-2.180 \times 10^{-4}) = 5.286 \times 10^4 \text{ J mol}^{-1}$ [1]; $E_a = +52.9 \text{ kJ mol}^{-1}$ [1].	2

Q	EXAMINER MARKING POINTS / WORKED MATHEMATICAL DERIVATIONS	MARKS
29(a)	Expt 1 to 2: [A] doubles, rate doubles ⇒ 1st order in A [1]; Expt 2 to 3: [B] doubles, rate quadruples ($9.6/2.4 = 4$) ⇒ 2nd order in B [1].	2
29(b)	Rate = $k[A][B]^2$ [1]; $k = (1.2 \times 10^{-3}) / ((0.10)(0.10)^2) = 1.2 \text{ mol}^{-2} \text{ dm}^6 \text{ s}^{-1}$ [1].	2
30(a)	Hydrogen gas escapes from the flask, causing a measurable decrease in total mass over time [1]; A cotton wool plug is placed in the neck to prevent loss of acid spray while allowing gas to escape [1].	2
30(b)	Draw a tangent to the curve at time $t = 60 \text{ s}$ [1]; Calculate the gradient of this tangent [1].	2
31(a)	$\text{NO}_2 + \text{CO} \rightarrow \text{NO} + \text{CO}_2$ [2].	2
31(b)	Rate = $k[\text{NO}_2]^2$ (since Step 1 is the slow step involving two NO_2 molecules) [1]; Intermediate = NO_3 [1].	2
32(a)	Reaction X is much faster [1]; Despite a lower A, its activation energy is much lower (50 vs 90 kJ mol^{-1}), making the $e^{(-E_a/RT)}$ factor vastly larger [1].	2
32(b)	Reaction Y shows greater sensitivity [1]; Reactions with higher activation energies experience a larger proportional increase in rate for a given temperature rise [1].	2
33(a)	The change in concentration of a reactant or product [1]; per unit time [1].	2
34(a)	The time taken for the concentration of a reactant to decrease to half of its initial (or original) value [2].	2
35(a)	The slowest step in a multi-step reaction mechanism [1]; which determines and limits the overall rate of reaction [1].	2
36(a)	The minimum amount of energy that colliding particles must possess [1]; in order for a chemical reaction to occur [1].	2
37(a)	A substance that increases the rate of a chemical reaction without being chemically changed or consumed at the end [1]; by providing an alternative reaction pathway of lower activation energy [1].	2
38(a)	Homogeneous: catalyst is in the same physical phase/state as the reactants [1]; Heterogeneous: catalyst is in a different physical phase/state from the reactants [1].	2
39(a)	The collision frequency factor / frequency of collisions between reactant particles [1]; taking into account the fraction of collisions with the correct steric orientation [1].	2
40(a)	The rate constant k remains unchanged [1]; k is only dependent on temperature (and the presence of a catalyst), not on concentration [1].	2
41(a)	Gradient = $-E_a / R$ ⇒ $E_a = -R \times (-1.25 \times 10^4) = 8.314 \times 12500 = 103925 \text{ J mol}^{-1}$ [1]; $E_a = +104 \text{ kJ mol}^{-1}$ [1].	2
41(b)	$\ln A = \ln k + (E_a / RT) = \ln(1.40 \times 10^{-3}) + (103925 / (8.314 \times 320)) = -6.571 + 39.06 = 32.49$ [1]; $A = e^{32.49} = 1.29 \times 10^{14} \text{ s}^{-1}$ [1].	2
41(c)	Units of A are s^{-1} [1]; In the Arrhenius equation $k = A e^{(-E_a/RT)}$, the exponential term is dimensionless, so A must have the exact same units as k (which is s^{-1} for a 1st order reaction) [1].	2
42(a)	Expt 1 to 2: $[\text{O}_2]$ doubles while $[\text{NO}]$ constant; rate doubles ⇒ 1st order in O_2 [1]; Expt 2 to 3: $[\text{NO}]$ triples while $[\text{O}_2]$ constant; rate increases 9-fold ($1.26 \times 10^{-2} / 1.40 \times 10^{-3} = 9 = 3^2$) ⇒ 2nd order in NO [1].	2
42(b)	Rate = $k[\text{NO}]^2[\text{O}_2]$ [1]; $k = (7.00 \times 10^{-4}) / ((0.010)^2 \times 0.010) = 7.00 \times 10^2 \text{ mol}^{-2} \text{ dm}^6 \text{ s}^{-1}$ [1].	2
42(c)	Rate = $(700) \times (0.025)^2 \times (0.015)$ [1]; Rate = $6.56 \times 10^{-3} \text{ mol dm}^{-3} \text{ s}^{-1}$ [1].	2
43(a)	1st half-life ($0.80 - 0.40$) = 40 s ; 2nd half-life ($0.40 - 0.20$) = 40 s ; 3rd half-life ($0.20 - 0.10$) = 40 s [1]; Successive half-lives are completely constant regardless of concentration, which is the defining signature of first-order kinetics [1].	2
43(b)	$k = \ln 2 / t_{1/2} = 0.693 / 40 \text{ s}$ [1]; $k = 0.0173 \text{ s}^{-1}$ [1].	2
43(c)	The reaction proceeds via an $\text{S}_{\text{N}}1$ mechanism where the slow, rate-determining step is heterolytic fission of the C-Cl bond forming a carbocation [1]; OH^- only attacks in the rapid second step after the rate-determining step, so its concentration does not affect overall rate [1].	2
44(a)	Step 1 has the higher activation energy barrier (TS_1 is the highest potential energy point on the path) [1]; Therefore, it has the smallest rate constant and is the slowest elementary step [1].	2

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44(b)	Since the rate is controlled by the slow step involving 1 molecule of NO ₂ and 1 molecule of F ₂ [1]; Rate = k[NO ₂][F ₂] [1].	2
44(c)	F is a reactive intermediate [1]; It is created during Step 1 and consumed in Step 2; a catalyst is present at the beginning and regenerated at the end [1].	2
45(a)	A downward sloping straight line with constant negative gradient terminating at the time axis [2].	2
45(b)	I ₂ does not participate in the rate-determining step (or any step prior to it) [1]; The slow step involves acid-catalysed enolisation of propanone; I ₂ only reacts in a fast subsequent step, so changing [I ₂] has no effect on rate [1].	2
46(a)	$k = \text{Rate} / ([A][B]) = (3.60 \times 10^{-4}) / (0.0500 \times 0.0800) = (3.60 \times 10^{-4}) / 0.00400$ [1]; $k = 0.0900 \text{ mol}^{-1} \text{ dm}^3 \text{ s}^{-1}$ [1].	2
46(b)	Units = $\text{mol}^{-1} \text{ dm}^3 \text{ s}^{-1}$ [1]; Reaction orders are determined experimentally by reaction kinetics and mechanism, not from the overall stoichiometric balancing coefficients [1].	2
47(a)	$\ln(1.00 \times 10^{-3} / 2.50 \times 10^{-4}) = \ln(4.00) = 1.386$ [1].	1
47(b)	$1/T_2 - 1/T_1 = 1/320 - 1/300 = -2.083 \times 10^{-4} \text{ K}^{-1}$ [1]; $E_a = -R \times 1.386 / (-2.083 \times 10^{-4}) = 8.314 \times 6653.9 = 55320 \text{ J mol}^{-1}$ [1]; $E_a = +55.3 \text{ kJ mol}^{-1}$ [1].	3
48(a)	Step 1: $2\text{Fe}^{2+} + \text{S}_2\text{O}_8^{2-} \rightarrow 2\text{Fe}^{3+} + 2\text{SO}_4^{2-}$ [1]; Step 2: $2\text{Fe}^{3+} + 2\text{I}^- \rightarrow 2\text{Fe}^{2+} + \text{I}_2$ [1].	2
48(b)	For Step 1: $E^\circ_{\text{cell}} = +2.01 - (+0.77) = +1.24 \text{ V} > 0$ (feasible) [1]; For Step 2: $E^\circ_{\text{cell}} = +0.77 - (+0.54) = +0.23 \text{ V} > 0$ (feasible) [1].	2
49(a)	Rate = k[A] [1]; $t_{1/2} = \ln 2 / k$ (or $t_{1/2} = 0.693 / k$) [1].	2
50(a)	An intermediate has fully formed chemical bonds and corresponds to a potential energy minimum [1]; A transition state has partially formed/broken bonds and corresponds to a potential energy maximum [1].	2